Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The most stable ion is

(A) $[Fc\,(C_2O_4)_3]^3$

(B) $[Fe(Cl)_6]^3$

(C) $[Fc\,(H_2O)_6]^{3+}$

(D) $[Fc(SCN)_6]^3$

Q.2

Which of the following is correct statement?

(A) Complex $[Co\,(H_2O)_6]^{2+}$ is more stable than $[Co\,(H_2O)_6]^{3+}$

(B) All complexes of $Ni(II)$(II) are bound to be outer d-orbital complexes.

(C) Stability constant for $[Ni(en)_3]^{2+}$ is greater than stability constant for $[Ni\,(NH_3)_6]^{2+}$

(D) Δ_{oct} for $[Cr\,(H_2O)_6]^{3+}$ is greater than Δ_{oct} for $[Cr\,(NH_3)_6]^{3+}$

Q.3

$C_{63}H_{88}CoN_{14}O_{14}P$ is the formulae of the Cyanocobalamine,
(vitamin B_{12}) it contain CN^- and CN^- is very poisonous, than why this compound does not prove to be fatal for us?

(A) CN^- forms covalent bond

(B) CN^- is coordinating to the cobalt as the ligand

(C) CN^- hydrolysis immediately

(D) All of these

Q.4

Which will give a white precipitate with $AgNO_3$ in aqueous solution

(A) $[Co\,(NH_3)_5\,Cl]\,(NO_2)_2$

(B) $[Pt\,(NH_3)_6]\,Cl_4$

(C) $[Pt(en)Cl_2]$

(D) $[Cu\,(NH_3)_4]\,SO_4$

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Q.5

For the reaction of the type $\mathrm{M} + 4\ \mathrm{L} \rightleftharpoons \mathrm{ML}_4$

(A) larger the stability constant, lower the proportion of ML_4 that exists in solution

(B) larger the stability constant, higher the proportion of ML_4 that exists in solution

(C) smaller the stability constant, higher the proportion of ML_4 that exists in solution

(D) None of the above

Q.6

Amongst $\mathrm{TiF}_6^{2-}, \mathrm{CoF}_6^{3-}, \mathrm{Cu_2Cl_2}$ and NiCl_4^{2-}, which are the colourless species?
(atomic number of $\mathrm{Ti} = 22, \mathrm{Co} = 27, \mathrm{Cu} = 29, \mathrm{Ni} = 28$)

(A) $\mathrm{Cu_2Cl_2}$ and $\mathrm{NiCl_4}^{2-}$

(B) TiF_6^{2-} and CoF_6^{3-}

(C) $\mathrm{CoF_6}^{3-}$ and NiCl_4^{2-}

(D) TiF_6^{2-} and $\mathrm{Cu_2Cl_2}$

Q.7

$\mathrm{InFe(CO)s}$, the Fe-C bond possesses

(A) \pi-character only

(B) Both σ and π characters

(C) lonic character

(D) σ -character only

Q.8

The crystal field splitting energy for octahedral $\left(\Delta_0\right)$ and tetrahedral $\left(\Delta_{\mathrm{t}}\right)$ complexes is related as

(A) $\Delta_{\mathrm{t}} = -\frac{1}{2}\Delta_0$

(B) $\Delta_{\mathrm{t}} = -\frac{4}{9}\Delta_0$

(C) $\Delta_{\mathrm{t}} = -\frac{3}{5}\Delta_0$

(D) $\Delta_{\mathrm{t}} = -\frac{2}{5}\Delta_0$

Q.9

Calculate the magnetic moment of Gd^{3+} ion $(\mathrm{Z} = 64)$.

(A) 7.95

(B) 9.73

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(C) 7.89

(D) 7.93

Q.10

For octahedral complex, which of the following d^n configurations of metal cation cannot exist in high spin and low spin forms?

(A) d^3 and d^8

(B) d^3 and d^5

(C) d^3 and d^6

(D) d^4 and d^8

Q.11

Which of the following complex ions has electrons that are symmetrically filled in both $_{2g}$ and e_g orbitals ?

(A) $[\mathrm{FeF_6}]^{3-}$

(B) $[\mathrm{Mn(CN)_6}]^{4}$

(C) $[\mathrm{CoF_6}]^{3-}$

(D) $[\mathrm{Co\,(NH_3)_6}]^{2+}$

Q.12

Which of the following statements related to crystal field splitting in octahedral coordination entities is incorrect?

(A) The $dx^2 - y^2$ and dz^2 orbitals has more energy as compared to d_{xy}, d_{yz} and d_{xz} orbitals.

(B) Crystal field splitting energy (Δ_{o}) depends directly on the charge of the metal ion and on the field produced by the ligand.

(C) In the presence of Br^- as a ligand the distribution of electrons for d^4 configuration will be $\mathrm{t_{2\,g}^3, e_g^1}$,

(D) In the presence of CN^- as a ligand $\Delta_{\mathrm{o}} < \mathrm{P}$.

Q.13

Among the following species the one which causes the highest CFSE, Δ_{o} as a ligand is:

(A) CN^-

(B) $\mathrm{NH_3}$

(C) F^-

(D) CO

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Q.14

Which one of the following complexes will most likely absorb visible light?
(At nos. $\mathrm{Sc}=21, \mathrm{Ti}=22, \mathrm{V}=23, \mathrm{Zn}=30$)

(A) $\left[\mathrm{Sc}\left(\mathrm{H}_2\mathrm{O}\right)_6\right]^{3+}$

(B) $\left[\mathrm{Ti}\left(\mathrm{NH}_3\right)_6\right]^{4+}$

(C) $\left[\mathrm{V}\left(\mathrm{NH}_3\right)_6\right]^{3+}$

(D) $\left[\mathrm{Zn}\left(\mathrm{NH}_3\right)_6\right]^{2+}$

Q.15

Incorrect match is :

(A) $\left[\mathrm{Rh}\left(\mathrm{PPh}_3\right)_3 \mathrm{Cl}\right]$: Wilkinson's catalyst

(B) $\left[\mathrm{Co}(\mathrm{CO})_4\right]^{-}$: Bond order of $\mathrm{Co}-\mathrm{CO}$ bond is greater than one

(C) $\left[\mathrm{Zn}\left(\mathrm{NH}_3\right)_4\right]\left[\mathrm{Be}(\mathrm{OH})_4\right]$: colorless complex

(D) $\left[\mathrm{Cr}(\mathrm{CN})_6\right]^3$: Inner orbital low spin complex

Q.16

Which of the following carbonyls will have the strongest $\mathrm{C}-\mathrm{O}$ bond ?

(A) $\left[\mathrm{Mn}(\mathrm{CO})_6\right]^{+}$

(B) $\left[\mathrm{Cr}(\mathrm{CO})_6\right]$

(C) $\left[\mathrm{V}(\mathrm{CO})_6\right]^{-}$

(D) $\left[\mathrm{Fe}(\mathrm{CO})_5\right]$

Q.17

Which of the following does not have a metal carbon bond?

(A) $\mathrm{Al}\left(\mathrm{OC}_2\mathrm{H}_5\right)_3$

(B) $\mathrm{C}_2\mathrm{H}_5\mathrm{MgBr}$

(C) $\mathrm{K}\left[\mathrm{Pt}\left(\mathrm{C}_2\mathrm{H}_4\right)\mathrm{Cl}_3\right]$

(D) $\mathrm{Ni}(\mathrm{CO})_4$

Q.18

In which of the following octahedral complex species the magnitude of Δ_o will be maximum?

(A) $\left[\mathrm{Co}\left(\mathrm{H}_2\mathrm{O}\right)_6\right]^{2+}$

(B) $\left[\mathrm{Co}(\mathrm{CN})_6\right]^{3-}$

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(C) $[Co(C_2O_4)_3]^{3-}$

(D) $[Co(NH_3)_6]^{3+}$

Q.19

The CFSE for $CoCl_6^{4}$ complex is 18000 cm^{-1}. The Δ for $CoCl_4^{2-}$ will be:

(A) 8000

(B) 9000

(C) 8100

(D) 7900

Q.20

Octahedral complex like $[ML_6]^{n+}$ has CFSE (Δ_o) of 1.8 eV. If identical metal ion (M^{n+}) forms a tetrahedral complex with the same ligands (L), then find CFSE of tetrahedral (Δ_t) of $[ML_6]^{n+}$ complex in electron volts (eV).

(A) 0.5

(B) 0.8

(C) 0.7

(D) 0.9

Answers & Solutions

Q.1 Answer:

$[Fc(C_2O_4)_3]^3$

Solution:

$$I^- < Br^- < SCN^- < Cl^- < S^{-2} < N_3^- < F^- < ONO^- < OH^- < SO_4^{-2} < NO_4^- < C_2O_4^{-2} < O^{-2} < H_2O \sim NCS^- < EDTA^{-4} < NH_3 \sim Py < en < bpy \sim phen < NO_2^- < PR_3 < CH_3 < CN^- \sim CO$$

From I^- To H_2O are weak field ligands

From NCS^- To CO are strong field ligands

C_2O_4 provides chelation to the atom, so it makes it more stable.

Q.2 Answer:

Stability constant for $[Ni(en)_3]^{2+}$ is greater than stability constant for $[Ni(NH_3)_6]^{2+}$

Solution:

(A) In $[Co(H_2O)_6]^{2+}$, the Co^{2+} (d^7) ion undergoes sp^3d^2 hybridization to form outer orbital complex. Thus the electronic arrangement of the metal ion will be $t_{2g}^6 e_g^1$.

In $[Co(H_2O)_6]^{3+}$, the Co^{3+} (d^6) ion undergoes d^2sp^3 hybridization to form inner orbital complex.

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Thus the electronic arrangement of the metal ion will be $t_{2\,g}^{6} e_{g}^{0}$.

$t_{2\,g}^{6} e_{g}^{0}$ is more stable than $t_{2\,g}^{6} e_{g}^{1}$.
Thus complex $\left[\mathrm{Co}\left(\mathrm{H}_2\mathrm{O}\right)_6\right]^{2+}$ is less stable than the complex $\left[\mathrm{Co}\left(\mathrm{H}_2\mathrm{O}\right)_6\right]^{3+}$.

Thus option A is incorrect.

(B) $\left[\mathrm{Ni}(\mathrm{CN})_4\right]^{2-}$ involves dsp^2 hybridization and is inner d-orbital complex.

Thus, all complexes of Ni (II) are not bound to be outer d-orbital complexes. Thus option (B) is incorrect.

(C) Ethylene diamine is chelating ligand and forms stable metal chelates whereas ammonia is unidentate ligand and does not form chelates.

Hence, Stability constant for $\left[\mathrm{Ni}(\mathrm{en})_3\right]^{2+}$ is greater than stability constant for $\left[\mathrm{Ni}\left(\mathrm{NH}_3\right)_6\right]^{2+}$.

Thus option C is correct.

(D) Water is weaker ligand when compared to ammonia.

Δ_{oct} for $\left[\mathrm{Cr}\left(\mathrm{H}_2\mathrm{O}\right)_6\right]^{3+}$ is smaller than Δ_{oct} for $\left[\mathrm{Cr}\left(\mathrm{NH}_3\right)_6\right]^{3+}$.

Thus option D is incorrect.

Q.3 Answer:

CN^{-} is coordinating to the cobalt as the ligand

Solution:

CN^{-} is coordinated to cobalt as the ligand and coordinated compounds have different properties than the individual species.

Q.4 Answer:

$\left[\mathrm{Pt}\left(\mathrm{NH}_3\right)_6\right] \mathrm{Cl}_4$

Solution:

$$\left[\mathrm{Pt}\left(\mathrm{NH}_3\right)_6\right] \mathrm{Cl}_4 \rightleftharpoons \left[\mathrm{Pt}\left(\mathrm{NH}_3\right)_6\right]^{4+} + 4\mathrm{Cl}^{-}$$
$$\mathrm{Ag}^{+} + \mathrm{Cl}^{-} \longrightarrow \mathrm{AgCl}(\mathrm{ppt})$$

Q.5 Answer:

larger the stability constant, higher the proportion of ML_4 that exists in solution

Solution:

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For the reaction of the type
$M + 4\,L \rightleftharpoons ML_4$**, larger the stability**
constant, the higher the proportion of ML_4 **that exists in solution.**

Q.6 Answer:

TiF_6^{2-} and Cu_2Cl_2

Solution:

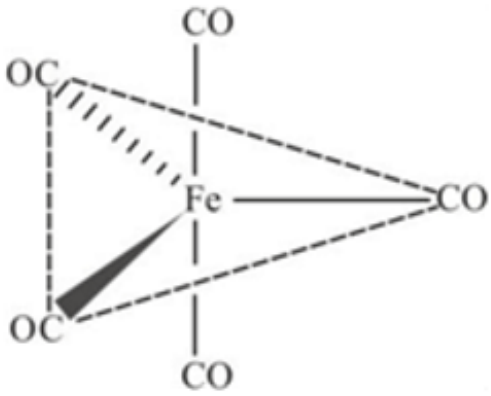
In $[TiF_6]^{2-}$ **and** $Cu_2Cl_2 \Rightarrow Ti^{4+}$ **and** Cu^{+}
Both ion do not have any unpaired electron in d orbital to (promote) transition. Hence no absorption of energy is there.
So they are colorless species.

Q.7 Answer:

Both σ and π characters

Solution:

there is $d\pi - p\pi$ **metal to ligand back bonding in Fe-C bond of the organometallic compound** $Fe(CO)_5$**. So it possesses both** σ **and** π **characters.**



Q.8 Answer:

$\Delta_t = -\frac{4}{9}\Delta_0$

Solution:

The crystal field splitting in tetrahedral complexes is lower than that in octahedral complexes, and $\Delta_t = -\frac{4}{9}\Delta_0$

Q.9 Answer:

7.93

Solution:

Gd $(64) = [Xe]4f^7 5d^1 6s^2$
$Gd^{3+} = [Xe]4f^7 5d^0 6s^0$

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i.e. no. of unpaired electrons $= 7$

$$\mu = \sqrt{n(n+2)} = \sqrt{7(7+2)}$$
$$= \sqrt{63} = 7.93\text{BM}$$

Q.10 Answer:

d^3 and d^8

Solution:

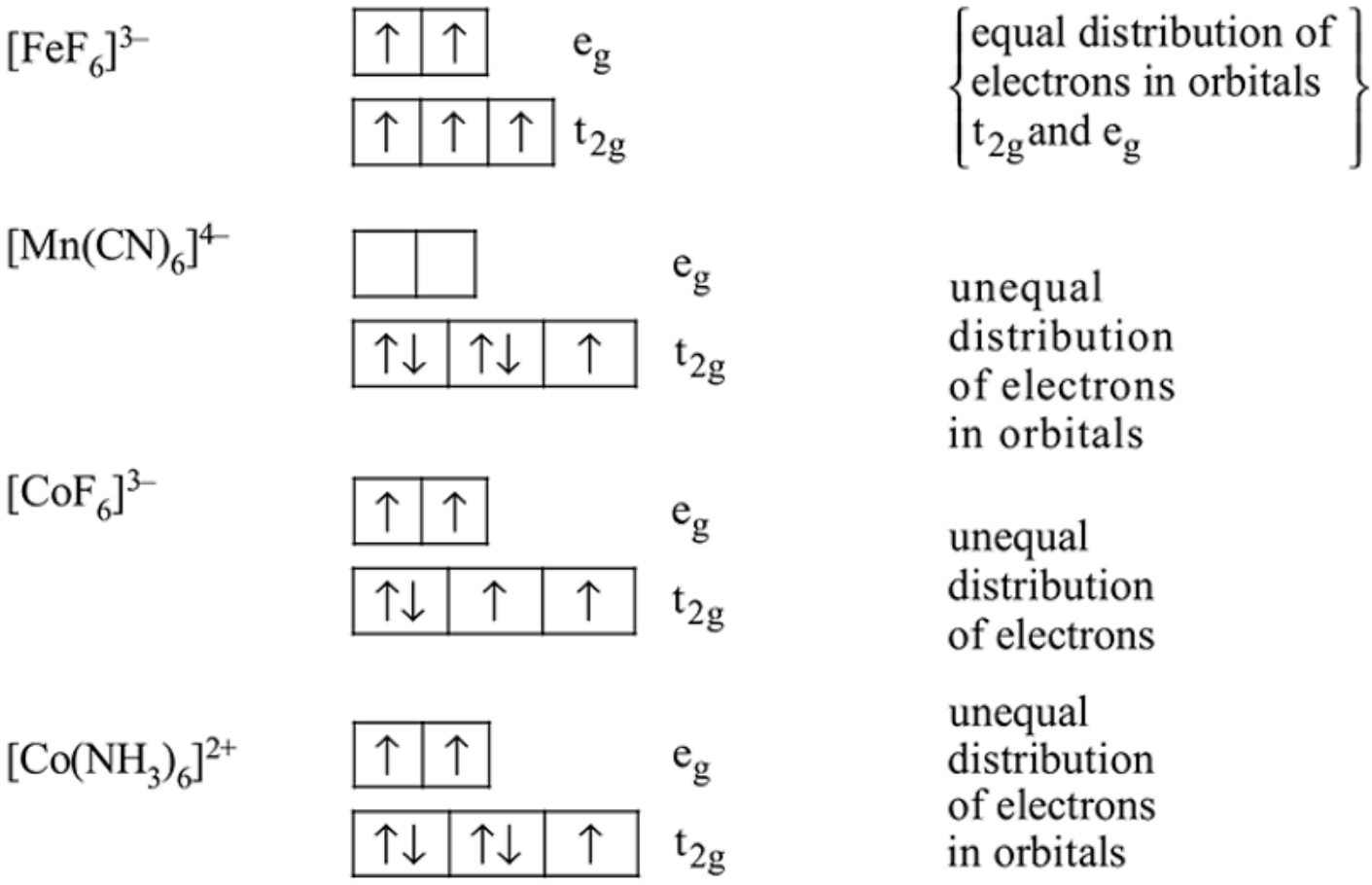
In d^3 and d^8 octahedral complexes number of unpaired e^-s at central metal atom/ion never changes, therefore for such octahedral complexes terms high spin and low spin not used.

Q.11 Answer:

$[FeF_6]^{3-}$

Solution:

Symmetrically filled t_{2g} and e_g are those, which contain equal distribution of electrons.



Q.12 Answer:

In the presence of CN^- as a ligand $\Delta_o < P$.

Solution:

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CN^- is a strong field ligand and form low spin complexes thus $\Delta_o > P$

Q.13 Answer:

CO

Solution:

Following is the order of increasing value of Δ_o for the ligands. $I^- < Br^- < Cl^- < F^- <$
$H_2O < Py = NH_3 < en <$
$NO_2^- << CN^- < CO$

Q.14 Answer:

$[V(NH_3)_6]^{3+}$

Solution:

The absorption of visible light and hence coloured nature of the transition metal cation is due to the promotion of one or more unpaired $-d$ - electron from a lower to higher level withing same d -subshell. Hence higher will be the number of unpaired electron higher will be the absorpion in visible light. The electronic configuration of the given elements is $Sc^{3+}(18) = 1s^2 2\ s^2 2p^6 3s^2 3p^6 3d^0 4s^0 -$ no unpaired e^-. $Ti^{4+}(18) = 1s^2 2\ s^2 2p^6 3s^2 3p^6 3d^0 4s^0 -$ no unpaired e^- $V^{3+}(20) = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^2 4s^0 -$ Two unpaired e^- $Zn^{2+}(28) = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^0 -$ no unpaired e^- hence $[V(NH_3)_6]^{3+}$ will most likely absorb visible light.

Q.15 Answer:

$[Cr(CN)_6]^3$: Inner orbital low spin complex

Solution:

$Cr(CN)_6]^{3-}$: Hybridization : d^2sp^3 High or low spin is not defined because Cr^{3+} is an example of d^3 system of configuration.

Q.16 Answer:

$[Mn(CO)_6]^+$

Solution:

As positive charge on the central metal atom increases, the less readily the metal can donate electron density into the π^* orbitals of CO ligand (donation of electron density into π^* orbitals of CO result in weakening of $C-O$ bond). Hence, the $C-O$ bond would be strongest in $[Mn(CO)_6]^+$

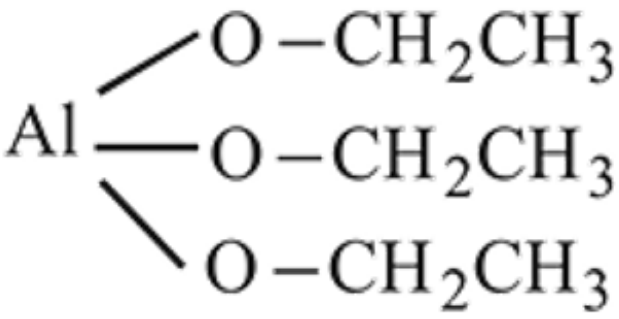
Q.17 Answer:

$Al(OC_2H_5)_3$

Solution:

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Triethoxyaluminium has no Al-C linkage



Q.18 Answer:

$[Co(CN)_6]^{3-}$

Solution:

Crystal field splitting depends upon the nature of ligand. The nature of ligand Δ decreases as shown below $C_2O_4^{2-} < H_2O < NH_3 < CN^-$

hence the crystal field splitting will be maximum for $[Co(CN)_6]^{3-}$

Q.19 Answer:

8000

Solution:

$\Delta t = \frac{4}{9}\Delta_0 = \frac{4}{9} \times 18000 = 8000$

Q.20 Answer:

0.8

Solution:

0.8

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